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ARTICLE

Phenotyping in clinical nutrition

Comparison of body composition assessment across body mass index categories by two multifrequency bioelectrical impedance analysis devices and dual-energy X-ray absorptiometry in clinical settings

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Abstract

Background InBody-770 and SECA mBCA 515 are multifrequency bioelectrical impedance analysis (BIA) devices, which are commonly used in the clinic to assess fat-free mass (FFM) and body fat (BF). However, the accuracy between devices in clinical settings, across different body mass index (BMI) groups remains unclear.

Methods Body composition for 226 participants (51% men, aged 18–80 years, BMI 18–56 kg/m²) was assessed by two commercial multifrequency BIA devices requiring standing position and using eight-contact electrodes, InBody 770 and SECA mBCA 515, and compared to results from dual-energy X-ray absorptiometry (DXA). Measurements were performed in a random order, after a 3 h fast and no prior exercise. Lin's-concordance correlation and Bland–Altman analyses were used to compare between devices, and linear regression to assess accuracy in BF% across BMI groups.

Results We found strong correlation between DXA results for study population BF% and those obtained by InBody ($\rho_c = 0.922$, 95% confidence interval (CI) 0.902, 0.938) and DXA and SECA ($\rho_c = 0.940$, CI 0.923, 0.935), with 95% limits of agreements between 2.6 and −8.9, and 7.1 and −7.6, respectively. BF% assessment by SECA was similar to DXA (−0.3%, $p = 0.267$), and underestimated by InBody (−3.1%, $p < 0.0001$). InBody deviations were largest among normal weight people and decreased with increasing BMI group, while SECA measurements remained unaffected.

Conclusions Both BIA devices agreed well with BF% assessment obtained by DXA. Unlike SECA, InBody underestimated BF% in both genders and was influenced by BMI categories. Therefore, in clinical settings, individual assessment of BF% should be taken with caution.

Introduction

In 2016, more than 1.9 billion adults worldwide were defined as overweight, and over 650 million as obese [1]. Among adults, a normal body mass index (BMI) is considered to be between 18.5 and 24.9 kg/m^2, with overweight defined as a BMI between 25.0 and 29.9 kg/m^2, and obese as a BMI of 30.0 kg/m^2 or higher. Obesity is further divided into class-1 or mild obesity (30.0–34.9 kg/m^2); class-2 or moderate obesity (35.0–39.9 kg/m^2), and class-3 or severe obesity (>40.0 kg/m^2) [2]. Although BMI is a practical tool to determine a healthy weight at the population level, its ability to assess adiposity and obesity-associated morbidities at the individual level is limited [1, 3, 4]. Body composition assessment can identify the main components of the human body, including fat mass (FM) and fat-free mass (FFM), which better associate with cardio-metabolic diseases than BMI alone [5, 6]. There are numerous established methods and techniques for in vivo estimation of body composition including bioelectrical impedance analysis (BIA), a widely used method for estimating FM and FFM [7]. BIA is based on measuring the resistance and reactance to an alternating current in the human body [8],

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where electrical conduction is related to the distribution of water and ions in the conductor. Because the FFM, which includes the protein matrix of adipose tissue, contains all the water and conducting electrolytes in the body, conductivity is far greater than in the FM [9]. BIA is a quick and relatively inexpensive method that does not require highly trained personnel, making it a practical and popular method in the clinical setting. The disadvantages of BIA mainly relate to the sensitivity to hydration status. The conductivity will therefore be proportional to total body water and to tissue with high water concentration (e.g., skeletal muscle). Furthermore, BIA devices are diverse, and rely on different technologies and equations [9]. Although the 4 compartment (4C) is consider as the gold standard for body composition assessment, dual-energy X-ray absorptiometry (DXA) is an accurate and precise method for differentiating lean and fat tissues, and is considered to be the reference method for body composition assessment [10, 11]. Recently, the results from two reviews highlighted the large variability in the ability of BIA predictive equations to obtain a similar value of FFM to that obtained by DXA, and revealed that the predictive equations are generally both population and device-specific [12, 13]. Previous studies suggested that BIA tends to underestimate FM and overestimate FFM relative to DXA, while multifrequency BIA (MF-BIA) appears to be a more accurate method for the assessment of body composition than single frequency [14, 15]. The accuracy was also found to be highly dependent on the suitability of the selected regression equation for a subject category, such as gender, age, or healthy versus unhealthy, within the population [9, 12]. A study [16] that compared the Medical Body Composition Analyzer (mBCA 514) SECA model to DXA, showed a strong agreement between the two methods, although it should be noted that this study involved overweight participants. Another study found strong correlation between DXA and InBody 770, although BIA overestimated whole-body muscle mass [17]. The accuracy of common MF-BIA devices to measure FM across BMI categories therefore remains unclear. The aim of this study was to assess the capability of two MF-BIA devices, to measure FM across different BMI categories in the clinical settings and compare the results to values obtained by DXA.

Materials and methods

Study population

Two hundred and thirty-seven men and women who attended the nutrition clinic were recruited into this cross-sectional study. Inclusion criteria were that participating adults should be between the age of 18 and 80 years with a BMI between 18 and 56 kg/m². Candidates with a defibrillator device and pregnant woman were excluded. The study has been approved by the ethic committee of Tel-Aviv University (0000607-2).

Anthropometric measurements

Weight (kg) was recorded to the nearest 0.1 kg on a digital scale (SECA model 400; SECA North America) while the subjects were barefoot and dressed in shorts and T-shirt. Height was measured to the nearest 0.5 cm by a SECA 274 Free-Standing Wireless 360 Stadiometer (SECA, Hamburg, Germany) and was used for both BIA devises. Waist circumference (WC) at the umbilicus was measured with measuring tape to the nearest 0.5 cm. Participants were instructed to exhale while standing, and a research assistant took two waist measurements. In the event of disagreement between the two measurements (≥2 cm) a third measurement was taken. BMI was calculated as weight divided by height squared (kg/m^2).

Body composition measurements

Participants were instructed to arrive at the clinic after at least a 3 h fast and having done no prior exercise that day. The Guidelines for DXA and BIA measurement are 8 h fast [18, 19], but since participants arrived at a different time of the day, including noon, they were asked to fast 3 h prior to the measurement, and not to abstain from breakfast in order to rehydrate and to fill glycogen content, as observed in previous works [20, 21]. For DXA, quality control calibration procedures were performed on a spine phantom each morning. All subjects had their body composition measured by DXA (Lunar; General Electric) and analyzed with the following software (enCORE 2011, ver. 13.60.; GE-Company, Madison, Wisconsin). Before the scan, subjects were asked to remove all metal items. Each participant performed a total body scan in a supine position to determine FM, lean body mass, and bone mineral density. Each whole-body scan took ~7 min. All subjects were measured by both the InBody 770 (InBody 770®) and SECA mBCA 515 (MFBIA; SECA®, Hamburg, Germany) BIA devices. For the InBody assessment, subjects stood barefoot on the platform of the device with the soles of their feet on the electrodes. They then grasped the handles of the unit with their thumb and fingers to maintain direct contact with the electrodes and remained still for ~1 min while keeping their elbows fully extended and their shoulder joint abducted to an ~30° angle. For SECA assessment, the subject stood on the device platform and grasped the grip rod as appropriate for their height. All measurements (height, weight, DXA, and MF-BIA) were taken while subjects were barefoot and dressed in shorts and T-shirt.

Table 1 Participants characteristics according to BMI-based subgroups.

	Normal weight		Over weight		Obese 1		Obese 2		All		*P*	*P*
	n = 102		*n* = 74		*n* = 35		*n* = 15		*n* = 226		Trend	ANOVA
Age (years)	38 ± 12		40 ± 13		45 ± 17		47 ± 13		40 ± 14		0.008	0.033
Sex (m, %)	41.2%		58.1%		60.0%		60.0%		50.9%		ns	
Weight (kg)	65.3 ± 8.2		79.6 ± 10.6		97.3 ± 11.4		114.0 ± 9.5		78.2 ± 18.1		<0.0001	<0.0001
BMI (kg/m^2)	22.7 ± 1.4		27.2 ± 1.4		32.3 ± 1.3		39.6 ± 5.1		26.8 ± 5.2		<0.0001	<0.0001
Waist circumference (cm)	84.5 ± 6.2		96.5 ± 6.9		111.1 ± 5.3		125.3 ± 12.2		95.4 ± 14.0		<0.0001	<0.0001
Neck circumference (cm)	34.2 ± 3.2		36.7 ± 3.3		40.3 ± 4.1		41.4 ± 4.1		36.6 ± 4.4		<0.0001	<0.0001
Body fat (%)	M	F	M	F	M	F	M	F	M	F		M F
DXA (%)	14.9 ± 4.0	31.3 ± 6.4	25.0 ± 6.2	38.9 ± 5.3	34.8 ± 4.0	46.2 ± 4.1	39.9 ± 6.9	48.5 ± 2.1	24.3 ± 9.9	36.2 ± 8.2	<0.0001	<0.0001 <0.0001
InBody (%)	13.1 ± 2.9	26.7 ± 5.7	22.1 ± 5.3	35.2 ± 5.7	31.9 ± 3.7	42.6 ± 2.6	39.7 ± 7.7	47.5 ± 2.8	22.0 ± 9.6	32.2 ± 8.6	<0.0001	<0.0001 <0.0001
Seca (%)	15.3 ± 4.3	30.3 ± 6.1	24.7 ± 5.5	39.6 ± 6.3	34.2 ± 3.2	45.0 ± 3.2	40.5 ± 6.3	48.6 ± 2.1	24.2 ± 9.6	35.7 ± 8.5	<0.0001	<0.0001 <0.0001

Values are means ± SD unless otherwise indicated. Linear regression was used to test the trend for anthropometric parameters and body fat percentage among BMI groups.

BMI body mass index, *DXA* dual-energy X-ray absorptiometry, *ns* not significant.

Measurement were performed in a random order with 30 s between measurements.

Test-related reliability for BF% and FFM measurements was determined by using 20 different subjects (not participants from this study) that were examined twice by DXA, SECA, and InBody. Intraclass correlation coefficients (ICC) and standard error of measurement (SEM) were determined for BF% (ICC > 0.994; SEM = 0.808%) and FFM (ICC > 0.962; SEM = 0.665 kg) for all devises. This was with agreement Vicente-Rodríguez et al. who concluded that for %BF measured with DXA, the intrarater, inter-rater, and interday TEM was lower than 0.5 U and $r > 99.7$ [22]. In a subset of 20 participants, body weight assessment was highly correlated between the body composition methods (ICC > 0.999, SEM = 0.404 kg).

Statistical analysis

Descriptive statistics are presented as mean ± standard deviation. Linear regression was used to test the trend for anthropometric parameters and body fat (BF) percentage among BMI groups. BF% values were tested for normality with Kolmogorov–Smirnov test. Two one-sided *t*-test was used to test equivalence. Lin's concordance correlation was used to assess the associations between devices. Bland–Altman method was used to assess level of agreement between methods. Paired samples *t*-test was used to test the between devices differences. An independent *t*-test was used to assess sex-related differences. Interaction between sex and BMI groups with the differences of the two devices as compared to the DXA, were performed using univariate model. Analyses were performed using SPSS Version 20.0 (SPSS, Inc., Chicago, IL, USA). All tests were two-tailed, and $p < 0.05$ was considered statistically significant.

Results

This study was designed to compare the ability of SECA mBCA 515 and InBody, two systems commonly used in clinical practice, to measure BF%, and to compare the results to those obtained with the reference method DXA. A total of 226 adults (115 men and 111 women) with a wide range of BMIs (18.73–55.4 kg/m^2) participated voluntarily in the study. Participants' anthropometric measures and BF % values from the different methods and systems are summarized in Table 1, according to BMI-based subgroups [2]. No significant differences in sex distribution were found between subgroups ($p = 0.110$). As expected, both WC and BF% increased with increasing BMI group. The values achieved by the reference method (DXA) and BIA systems indicated a significant correlation between DXA

and InBody ($\rho_c = 0.922$, 95% confidence interval (CI) 0.902, 0.938) and DXA and SECA ($\rho_c = 0.940$, CI 0.923, 0.935) for the entire cohort, and within sex groups (Fig. 1A, B). These results indicate that at the level of the population, there is strong correlation between both BIA systems and the reference method.

Next, we compared both InBody and SECA to DXA using Bland–Altman analysis. InBody values were significantly different to those obtained with DXA (Fig. 2A), with a mean underestimation of $-3.1 \pm 2.9\%$ and 95% LOA of -8.9%, 2.6%, with a larger underestimation for women (bias: $-4.0 \pm 2.8\%$, 95% LOA: -9.5%, 1.5%) than men (bias: $-2.3 \pm 2.8\%$, 95% LOA: -7.8%, 3.2%) (Fig. 2B). Although SECA values were not significantly different (-0.2%, $p = 0.267$), they exhibit (Fig. 2C) a wide limit of agreement (-7.6%, 7.1%, ±3.7), particularly in women (Fig. 2D). Similar trends were observed when we compare the ability of the two devices in measuring FFM. Values obtained by InBody had a bias of 3.1% ($p < 0.0001$) as compared to those obtained by DXA (Fig. S1A), with a greater bias (4%) among female (Fig. S1B). As for BF%, the FFM% values obtained by SECA were not significantly different (0.3%, $p = 0.267$) than those obtained by DXA, but with a wide LOA (-7.0%, 7.6%, ±3.8).

According to the American College of Sports Medicine (ACSM) [23], difference from DXA between 2 and 5% are considered accurate. We therefore tested for equivalence using a range of ±3% ±5% differences. Both InBody and SECA were statistically equivalent ($p < 0.0001$). We next plotted the absolute differences from DXA obtained by InBody and SECA in order to assess the distribution of deviation for each system (Fig. 3). The results indicated that

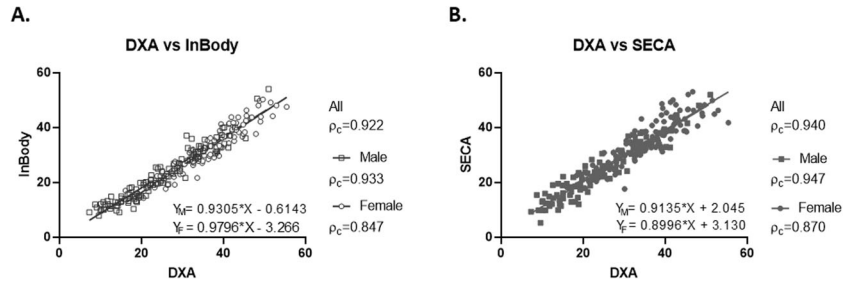

Fig. 1 Comparison of body fat between multifrequency bioelectrical impedance analysis (BIA) devices with DXA. Lin's-concordance correlation (**A**) DXA and InBody and (**B**) DXA and SECA for the entire cohort, and within sex groups (Male—red balls, Female—blue balls).

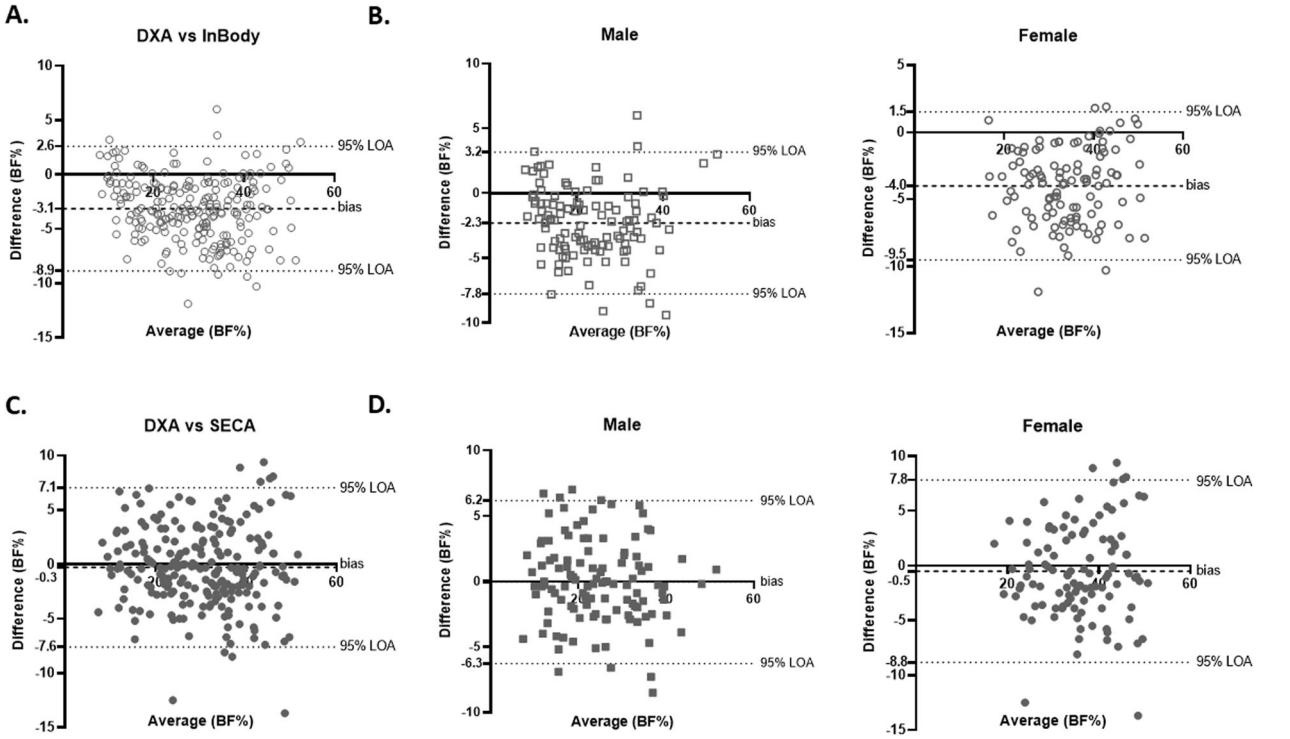

Fig. 2 Blant–Altman analysis for BF% differences from DXA, for both InBody and SECA. A Bland–Altman analysis between DXA and InBody for the entire cohort and (**B**) within sex groups (Male—red circles, Female—blue circles). **C** Bland–Altman analysis between DXA and SECA for the entire cohort and (**D**) within sex groups (Male—red balls, Female—blue balls). Dotted line—95% limit of agreement (LOA). Dashed line—bias.

23% of the measurements obtained with SECA were within ±1% deviation, compared to only about 17% InBody values. While the percentage of measurements within ±3% deviation of BF% from DXA were equally distributed for SECA (−19.9% and +20.3%), the differences for InBody were an underestimation (−54.4% and +1.8%). In addition, 25.2% of the measurements for InBody and 8.8% for SECA were >−5% BF, while 0.4% and 9.7% of the measurements, respectively, were >5% BF. In total, most measurements in both systems fell within the accountable range of ±5% from the DXA measurements, with ~26% outside the range for InBody (mainly underestimation) and 18% for SECA.

Considering the deviations, we asked whether there was any relationship between the differences in BF% produced by the different methods and the BMI groups. To this end, we investigated the deviations from DXA within the different clinically used BMI categories (Fig. 4). The results indicated that the SECA measurements were not significantly affected by BMI, while the InBody deviations were largest among normal weight people and decreased with increasing BMI group, until there was no significant difference from the DXA results found among people in obese-2 group. In addition, trend of interaction ($p = 0.102$) between sex and BMI was found for the InBody differences, with nonsignificant interaction for the effect on SECA differences ($p = 0.388$).

Discussion

In the present study, we compared the assessment of BF% values by two MF-BIA devices (SECA mBCA 515 and InBody 770), against DXA as a reference. Our results indicated a strong correlation between BF% measured by DXA and SECA or InBody, although the values of BF% were underestimated by InBody in both genders, and to a greater extent in females. This underestimation was more pronounced in the lower BMI categories, but not in the obese-2 class. In contrast, the average values for the whole population achieved by SECA were not significantly different from the reference values, but with a wider limit of agreement. Our results reveal that although the values obtained by the BIA methods are very comparable to the DXA results, the two devices tested have different patterns of BF% misassessment.

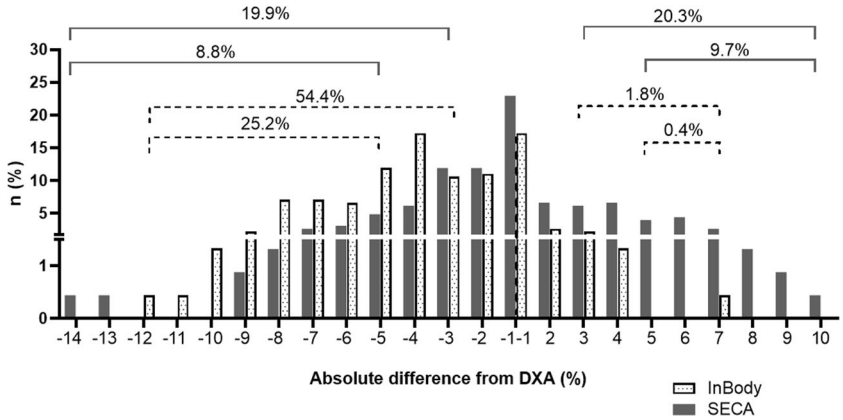

Fig. 3 Distribution of absolute differences of BF% obtained by InBody (▭) and SECA (■) from DXA. Lines in the figure represent the percentage of measurements falling within ±3% and ±5% of the DXA value for InBody (dashed line) and SECA (gray line).

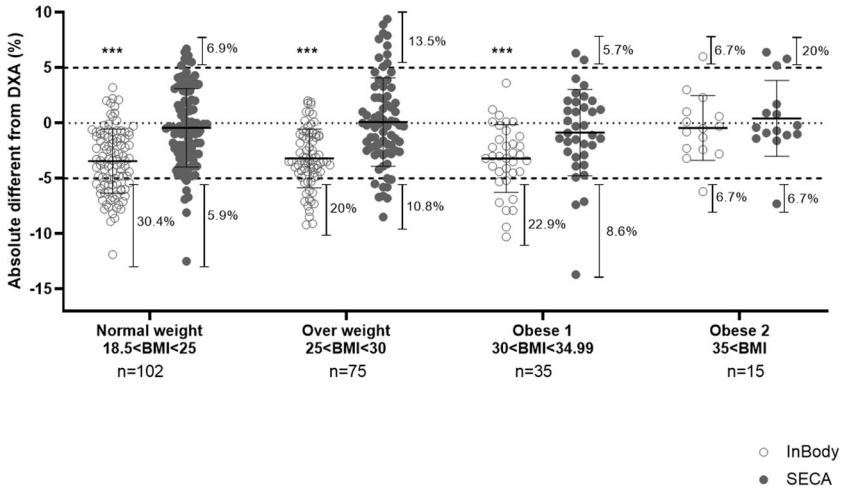

Fig. 4 Comparison of the absolute difference of BF% obtained by InBody (gray circles) and SECA (gray balls) from DXA, within different clinically used BMI categories. Lines represent the percentage of measurements falling within ±5% of the DXA value. Paired samples *t*-test was used for between methods comparison. ***$p < 0.001$.

The major finding of this study is the wide diversity of agreement in assessing BF%, ranging from −11.9 to 6% for InBody, and −13.7 to 9.4% for SECA, which queries the precision of the individual BF% estimations. These results are in agreement with a previous cross-sectional study that compare the assessment of body composition among 67 man and women, by three different InBody devices (InBody230, InBody720, and InBody770) as compare to DXA. The authors found a systemic underestimation of both FM and BF%. Unlike the results from our study, the authors did not find sex differences. Since the underestimation was <3.3% [24], the author concluded that all devices could be used for body composition assessment at the individual level, when DXA is not available [24]. Other cross-sectional studies of BF% reported strong correlations between DXA and three different BIA devices, in this study the author conclude, whereas that MF-BIA demonstrated the strongest correlation with DXA, it provided the poorest absolute level of agreement, with a large bias in underestimating BF content (by ~7%) [21]. Our results are in accordance with a recent study by Fonseca et al. which investigated the validity of an eight-contact electrode BIA system within a household scale, for assessing whole-body composition in 17 COPD patients. Although there was a strong inter-method correlation for FM ($r = 0.9$), the results showed discrepancies in FM, ranging from −6.71 to 5.61 kg [25]. Although the average deviation was <3.3%, it ranged between +2.3% and −9.4% (for InBody770). The precision of most methods can be considered high if well-trained persons perform the measurements and appropriate models that translate body measurements to body composition variables are used. Comparative studies between different multicomponent methods showed a standard error of the estimate for BF% <1% [26] and mean biases between 1% and 2.1% BF [27]. Accuracy for field methods can range between 3 and 4% deviation from 4C, MRI, or CT [23]. Although the accuracy of predicting FM and FFM depends on the method, 2 to 5% deviation are considered accurate [28]. As suggested by Ward [29], clinicians and clinical researchers may be more interest in what is considered "important" or "worthwhile" in practice. Unfortunately, these are not well defined, particularly in body composition research, where minimal clinical important difference, is not well establish. Considering all off the above, we choose 3 and 5% as clinically acceptable range. As mentioned, the ACSM guide for body composition assessment, defined clinically acceptable equivalence smaller than 2–5% deviation from the reference method, and values above that are considered poor. In the current study, the average bias for InBody was indeed −3.1%, but overall, 24.8% of all BF% measurements were an underestimation >5%. These underestimations were significant in the normal weight, overweight and obese 1 group.

On the other hand, we found that SECA mBCA 515 assessments of BF% were not statistically different than the values achieved by DXA (−0.2%, $p = 0.267$) but with a wide limit of agreement (−7.6%, 7.1%, ±3.7). Therefore, for the population, the %BF measurements obtained by SECA are in good agreement with the reference method, but for the individual, wide LOA may put the results in caution. These results are in agreement with a recent paper that used SECA for body composition assessment as compare to DXA [10]. The authors found that FM assessed by SECA showed great agreement with the values taken by DXA, particularly among people that are overweight and with high amount of FM. In contrast, the authors also found that clinically acceptable equivalence (deviation <5%) was only achieved in 25.8% of the cases. In this study with a smaller sample size (30 adult participants), comparing DXA to BIA (SECA 515) across the spectrum of BMI, have shown wide level of agreement of FM range from −3.8 to 4.4 kg, and overestimated FFM—8.2 to 4.3 kg [10]. In our measurements, 82.3% of all SECA's BF% assessments deviated <5%, and 23% from all measurements ranged between ±1% deviation from DXA. Nevertheless, in 17.7% of all SECA's assessments were not clinically equivalence, reaching deviation of up to −14%. These extreme values were across all BMI categories.

Our results here indicate that although the correlations between BIA methods (SECA and InBody) and DXA are good and comparable, the two devices have different patterns of BF% misassessment, which are differently affected by BMI. We found that SECA results were not affected by BMI. SECA results were in good agreements with DXA in 23% of all measurements ranged between ±1% of the corresponding DXA value. In patients with a normal BMI ($18.5 < \text{BMI} < 25$), SECA overestimated BF% by more than a 5% difference from the DXA value in 6.9% of subjects, this rose to 13.5% of overweight patients, and then dropped to 5.7% of obese patients. An underestimation of more than 5% was found in 5.9%, 10.8%, and 8.6% of normal BMI, overweight, and obese subjects respectively. Altogether, the SECA results showed a relatively high agreement with DXA values. This is not in agreement with previous studies that found that three BIA devices underestimated values in overweight and obese men compared to DXA [21]. The authors concluded that compared with DXA, MF-BIA produced a large bias with wide limits of agreement, and that the accuracy of estimating body composition in overweight or obese men was poor ($r^2 = 0.69$–0.4). Another similar study suggested that concordance between BIA and DXA methods at the individual level was lacking, irrespective of BMI [30]. The results of BF% in overweight ($\text{BMI} > 25$) and obese ($\text{BMI} > 30$) Colombian Adults, indicated that mBCA 514 and DXA are comparable methods for measuring body composition with higher BF percentages ($r = 0.921$, $p = 0.001$). The author of this study concluded that females with a $\text{BMI} > 30$ had a higher

correlation ($r = 0.970$, $p < 0.001$), than males (BMI > 30), although the correlation for men was also fairly high ($r = 0.868$, $p < 0.001$) [16]. In our study the correlation between DXA and SECA for males and females with BMI > 30 was $r = 0.850$ ($p < 0.0001$). When comparing SECA 515 to DXA, Day et al. reported that overweight and obese groups showed narrower limits of agreement than the normal weight group [10]. In our measurements, 81.3% of all SECA BF% assessments deviated <5%, and 61% of all measurements were between ±1% of the DXA value. Nevertheless, 18.7% of all SECA assessments were not clinically equivalent, reaching deviations of up to −14% from the reference DXA values. These extreme values were present across all BMI categories.

This study has several limitations. This study aimed to assess the capability of two specific MF-BIA devices (InBody 770® and SECA mBCA 515) to measure FM across different BMI categories in the clinical settings. Therefore, the results above are only relevant for these two particular devices and their built-in algorithms, and not the BIA technique in general. In addition, BIA methods are affected by the hydration status, which was not regulated in the current study and therefore may affect the results. However, all measurements were done at the same time, with only a few minutes separating the measurements, which were done in a random order by the same technician. It should be noted that this sort of scenario, where the assessments are made at different time-points throughout the day, mimics the clinical settings better than standardized morning measurements. Another limitation is the relatively small sample size of subjects with BMI > 35. However, the relatively high subject number and the large range of BMIs in this study was a strength and allowed an assessment of BF% over different ranges of BMI. Another strength is the wide spectrum of participants' age, ranging from 18 to 77 years, which together with the wide range of BMI, makes the study relevant for most people.

In conclusion, our finding suggests that although both BIA methods were highly correlated with DXA, at the individual level, InBody underestimated BF% to a higher degree then SECA in both genders and in all BMI categories. Although at the population level, both devices have an acceptable bias, the wide LOA may put the measurements of an individual in caution. These results may affect patient treatment in the clinic, and therefore, should be taken into consideration in a clinical setting.

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Author contributions Conceptualization, YL, NG, and YG; methodology, YL and YG; formal analysis, NG; investigation, YL and NG; resources, YL and YG; data curation, YL and NG; writing—original draft preparation, YL, NG, and YG; writing—review and editing, YG; visualization, NG; supervision, YG; project administration, YL.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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